Pawpaw-ssible Solutions to Rural Food Insecurity

Geospatial Viability Analysis on Pawpaw Growth within Rural Food Deserts

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Introduction & Background



Image 1. Ripe Pawpaw

Over 17 million U.S. households faced food insecurity in 2022, with rural areas disproportionately affected due to limited access to fresh, nutritious food (USDA, 2021). Traditional responses—such as grocery store incentives and urban agriculture—often fall short in rural contexts due to infrastructure gaps and geographic isolation.

This study investigates the potential of pawpaw (*Asimina triloba*), a native, nutrient-dense fruit tree, as a sustainable intervention in rural food deserts. The pawpaw thrives in varied environmental conditions, requires minimal agricultural inputs, and holds deep cultural significance in Appalachian, Indigenous, and African American communities.

Using Geographic Information Systems (GIS), we evaluate ecological viability for pawpaw cultivation in two USDA-defined 20-mile rural food deserts: Highland County, Virginia, and McCurtain County, Oklahoma. By integrating environmental variables (e.g., slope, Solar irradiance, soil pH) with food access data, this research offers a replicable spatial planning framework to support hyper-local, climate-resilient food solutions.



Image 2. Fully grown Pawpaw on the tree

Methods & Data

Locating Area of Interest

For this step, tabular data was merged with spatial vectors in **ArcGIS** to insert the necessary information with their visual representations.

Tabular Data

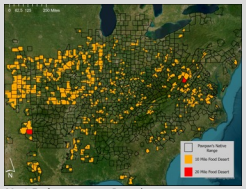
- **(USDA)** Pawpaw County Distribution
- **(USDA)** Food Access Research Atlas

Important Variables

- Rural Designation
- 10- or 20-Mile Food Desert Designation
- Poverty Rate (%)

Vector Data

- (**ESRI**) Open-Source County Polygon
  - Merged with Pawpaw Distribution Data
- (**ESRI**) Open-Source Census-Tract Polygon
  - Merged with Food Access Research Atlas Data



Map 1. Food insecurity across Pawpaw's native range

Environmental Suitability

Methods and Analysis

Data

For this section of analysis environmental data was represented by raster spatial data cells. Rasters are a method of representing geographic information using a grid of equally sized cells that represents a specific attribute of the area it covers.

Variable	Source	Weight
Soil pH	ESRI Living Atlas	1
Land Use	Commission for Environmental Cooperation	1.5
Precipitation	PRISM climate Group at Oregon State University	1
Solar Irradiance	Generated using 30sec DEM from HydroSHEDS.org	1
Slope	Generated using 30sec DEM from HydroSHEDS.org	1

Weighted Layer Overlay Model

Weighted raster overlay is used to evaluate land suitability by combining multiple environmental variables into a single map. In this study, raster layers were standardized on a 1–3 scale and weighted based on their benefit to pawpaw growth. Variables are given a multiplier to highlight the importance of specific layers . Land use was the only layer given a **1.5** multiplier because of its necessity to eliminate non-arable areas. The layers are transformed, weighted, and combined to highlight areas most viable for cultivati

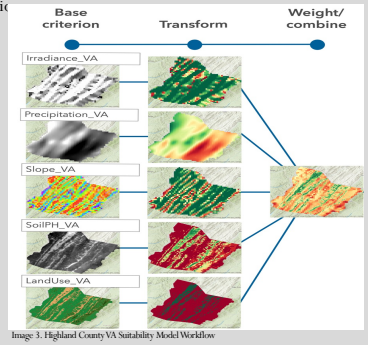
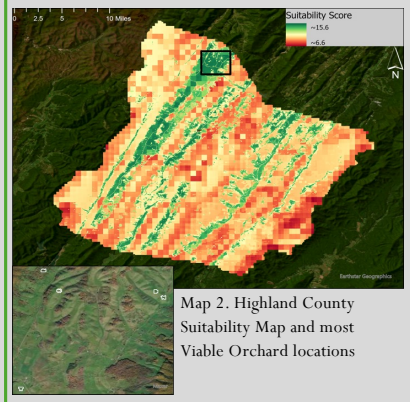


Image 3. Highland County VA Suitability Model Workflow

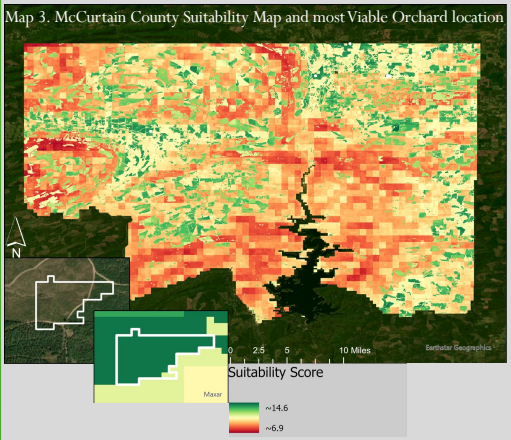
Results and Discussion



Map 2. Highland County Suitability Map and most Viable Orchard locations

Highland County, designated by the USDA as a 20-mile food desert (U.S. Department of Agriculture, 2023), contains multiple high-suitability areas for pawpaw cultivation. Each 4-acre site identified in the model can support approximately 1,180 trees, based on a planting density of 295 trees per acre (University of Kentucky, 2012). With a projected gross return of $9,600 per acre at a wholesale price of $1.60 per pound, this translates to an estimated 6,000 pounds of fruit per acre—or **24,000 pounds of fruit per 4-acre orchard** at full maturity. These high-yielding sites align with regions of Highland County where 25% of households earn less than $35,000 annually, and over 20% of residents are aged 65 or older (Esri, 2020). Suitability scores for the selected locations ranged from 15.3 to 15.5, reflecting strong alignment between environmental conditions and productive potential. These results indicate that pawpaw cultivation could offer a viable strategy to increase access to fresh produce while supporting local economic resilience in an aging, rural population.

Map 3. McCurtain County Suitability Map and most Viable Orchard location



McCurtain County, Oklahoma is designated by the USDA as a 20-mile food desert (U.S. Department of Agriculture, 2023), underscoring the importance of local food production initiatives. Suitability analysis identified multiple viable orchard locations, including a standout 16-acre plot with an average suitability score of 14.1. At the recommended planting density of 295 trees per acre, this site could support approximately **4,720 pawpaw trees** (University of Kentucky, 2012). Based on a wholesale price estimate of $1.60 per pound, the University of Kentucky study projects gross returns of $9,600 per acre, translating to a potential yield of **96,000 pounds of fruit** for this 16-acre site. This production capacity is particularly significant given that 15.1% of McCurtain County households earn under $15,000 and the local unemployment rate is 5.1%, more than double that of Highland County (Esri, 2020). These findings suggest that pawpaw cultivation in McCurtain County could support both food security and economic development in a region marked by persistent rural poverty and limited fresh food access.

Highland County Suitable Locations

Raster Cell Count	Area (Acres)	Avg Suitability Score	Total Suitability Score
18	4	15.5	277.9
18	4	15.4	277
18	4	15.3	276.2
18	4	15.3	275.8
18	4	15.3	275.7

McCurtain County Suitable Locations

Raster Cell Count	Area (Acres)	Avg Suitability Score	Total Suitability Score
5	1	14.4	71.8
72	16	14.1	1020.4
5	1	14	70.3
5	1	13.9	69.9
5	1	13.9	69.7

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Citations

- Esri. (2020). *Highland County Key Facts* [Infographic]. Esri.
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